Pseudo first order reaction:

Kinetic study of a higher order reaction is difficult to follow, for example, in a study of a second order reaction involving two different reactants; the simultaneous measurement of change in the concentration of both the reactants is very difficult. To overcome such difficulties, A second order reaction can be altered to a first order reaction by taking one of the reactant in large excess, such reaction is called pseudo first order reaction. Let us consider the acid hydrolysis of an ester,

$$CH_3COOCH_3\ (aq) + H_2O\ (l) \xrightarrow{H^+} CH_3COOH\ (aq) + CH_3OH\ (aq)$$

$$\text{Rate} = k\ [CH_3COOCH_3]\ [H_2O]$$

If the reaction is carried out with the large excess of water, there is no significant change in the concentration of water during hydrolysis. i.e.,concentration of water remains almost a constant.

Now, we can define $k\ [H_2O] = k'$; Therefore the above rate equation becomes

$$\text{Rate} = k'\ [CH_3COOCH_3]$$

Thus it follows first order kinetics.

7.5.2 Integrated rate law for a zero order reaction:

A reaction in which the rate is independent of the concentration of the reactant over a wide range of concentrations is called as zero order reactions. Such reactions are rare. Let us consider the following hypothetical zero order reaction.

$A \longrightarrow$ product

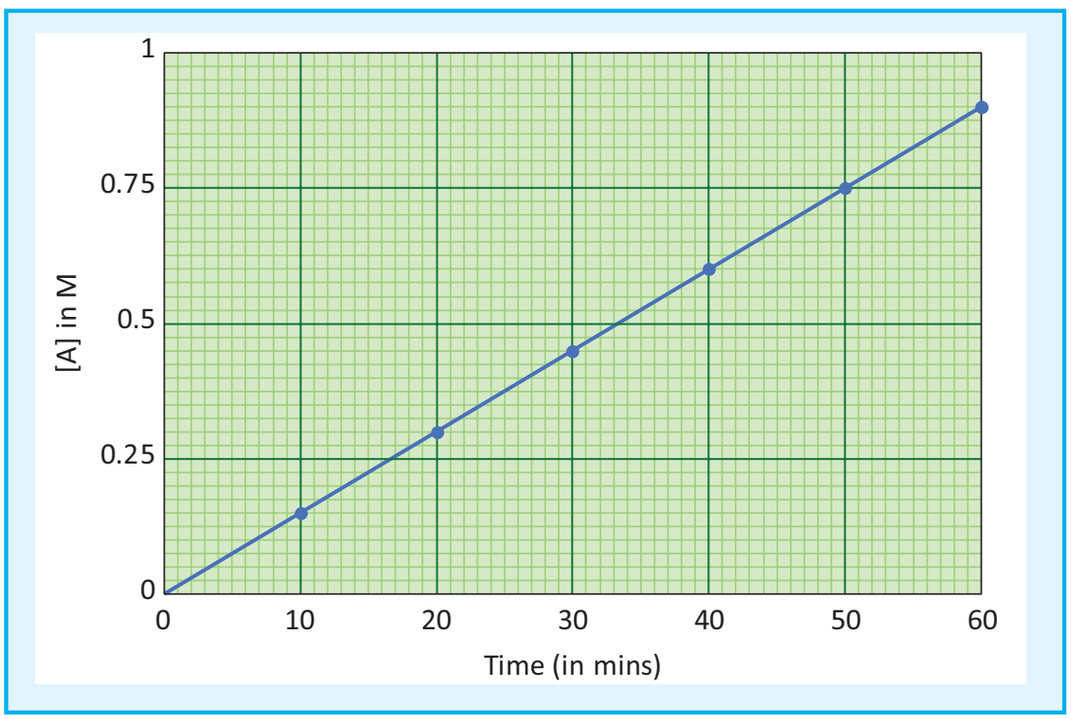


Fig 7.4 : A plot of [A] Vs time for a zero order reaction $A \longrightarrow$ product with initial concentration of [A] = 0.5M and $k = 1.5 \times 10^{-2} mol^{-1} L^{-1} min^{-1}$

XII U7 kinetics - Jerald Folder.indd 214 3/29/2019 6:34:37 PM